WORKING BLUEPRINT · V1.0 · 13 NOV 2026

The Forward Deployed Engineer course: content pipeline blueprint

How we build a premium Udemy and YouTube course system that follows the Forward Deployed Engineer skill roadmap, with a repeatable, high-quality, affordable production pipeline. This is the document we start working from.

Human-led · AI-accelerated · Code-verified

Prepared for: course owner

Status: draft for decisions (see §11)

01 Executive summary

We'll build an original course family that teaches the skills on the Forward Deployed Engineer (FDE) roadmap, sold on Udemy and fed by a YouTube channel. The product is **the pipeline**: one structured lesson script turns into a 16:9 course lesson, 9:16 Shorts, captions, a thumbnail and metadata, with automated quality checks.

~8×
growth in US FDE job postings, Apr 2025 → Apr 2026 (Indeed: 643 → 5,330)

$174–188k
median advertised FDE base salary (Bloomberry / Lightcast)

≈4 h
typical length of existing Udemy FDE courses; most are new with few reviews

The bet

Demand for FDEs is surging, and existing courses are short, generic or exam-prep. Our edge is **production quality plus real engineering**: animated, code-verified lessons, a full customer-engagement capstone, and a consistent brand across YouTube and Udemy.

Key recommendations

1. **Follow the roadmap and never republish it.** roadmap.sh content is copyrighted (personal use only). We cover the same skill areas with our own curriculum, structure, wording and visuals, and link to roadmap.sh as a free companion resource.
2. **Human-led, AI-accelerated.** Udemy flags fully AI-generated courses with little instructor input, and YouTube demonetizes template-based, repetitive "inauthentic" content. So the instructor's real voice (recorded, or a verified clone of their own voice), real judgment and real code are in every lesson.
3. **Ship a focused first course**, not the whole roadmap: *"AI Engineering for Forward Deployed Engineers"* (≈8–10 h). It covers the highest-demand, least-served part of the roadmap. Foundations courses come later.
4. **Build the pipeline on Remotion** (React video framework, free for individuals and companies of up to 3 people), with ElevenLabs voice plus word timestamps, VHS for terminal demos, and CI that runs every code sample.
5. **Lean budget: ≈ $50–150/month** during production, mostly voice credits. No expensive SaaS video editors needed.

> **First 2 weeks, in one line**
>
> Fix the voice setup, lock the brand name, create the repo and channel, build a pipeline MVP with 5 scene templates, ship 1 pilot lesson and 3 Shorts, and send Udemy's free test video for feedback.

02 Opportunity and competition

Market signal

- **Demand:** Indeed data cited by Business Insider shows US FDE postings rising from 643 (Apr 2025) to 5,330 (Apr 2026). Bloomberry reports +1,165% YoY across 1,000 postings, and Paraform reports 10× growth in 18 months. India grew ~19× in a year.
- **Pay:** median base ≈ $174k (Bloomberry), median advertised salary > $188k vs ≈ $145k for a traditional SWE role (Lightcast). Palantir FDE median total comp ≈ $215k (Levels.fyi, May 2026).
- **Why:** AI vendors (OpenAI, Anthropic, Palantir and many startups) need engineers who can deploy AI inside customer systems. That takes engineering plus discovery, integration and stakeholder skills, a mix few people are trained in.

Competitive landscape (Nov 2026)

Offer	Shape	Gap we exploit
~8+ Udemy FDE courses (e.g. "FDE Mastery", "FDE 2026", "Complete Crash Course", CFDE prep, practice tests)	Mostly 2–5 h, slide or talking-head, recently launched, few ratings (some under 3★)	Depth, production quality, real runnable projects, regular updates
Maven cohorts and live trainings	Live, $500–$2,000+, time-bound	Self-paced, affordable, always available
Educative / fde.academy	Text-first or certification-style	Video-first, visual explanations, YouTube reach
Free YouTube content	Scattered career talk, few structured build-alongs	A structured, branded series that funnels into the paid course

> **Platform context**
>
> Udemy and Coursera agreed to merge (closing expected H2 2026). Marketplace dynamics may shift, so the plan keeps an **owned audience** (YouTube + email list) so we're never dependent on one storefront.

03 Rules we must design around

These constraints shape the whole pipeline. Breaking them would cost us reach, revenue or the right to publish.

roadmap.sh: follow it, don't publish it MUST

- The roadmap's license allows **personal use only**. Publishing its text, images or files in any form needs the copyright holder's consent. Linking to roadmap.sh is allowed.
- **Our approach:** use the roadmap privately as a *coverage checklist* (its ~100 topic areas, §5). We write our own curriculum, module structure, explanations, diagrams and examples, and never reproduce the roadmap diagram, its descriptions or its resource lists.
- Don't use "roadmap.sh" in the course title or brand, or imply endorsement. A line like "pairs well with the free FDE roadmap at roadmap.sh" with a link is fine.
- OPTIONAL Contact the roadmap author about a partnership (e.g. being listed as a course resource). That's upside, not a requirement.

Udemy MUST

- **Minimums:** ≥ 30 min of video, ≥ 5 lectures, ≥ 720p, 16:9, clear audio in both channels, in sync. Use Udemy's free **test video** review before full production.
- **AI:** AI-assisted production is allowed, but Udemy warns against fully AI-generated courses with little instructor involvement, factual errors and low-quality AI audio or visuals. **Disclose the AI tools used in the course description.** Udemy Business inclusion needs content that has been manually reviewed and polished.
- **Revenue:** marketplace sales without instructor promotion use a single blended rate (it replaced separate organic and ads rates; check the current %). Instructor-coupon sales keep ~97%. The subscription pool share dropped to 15% in Jan 2026. **So our own funnel (YouTube → coupon) is worth far more per sale.**

YouTube MUST

- The **"inauthentic content"** monetization policy (clarified July 2025, expanded July 2026) targets mass-produced, repetitive, template-based videos with little variation, *whether or not* AI was used. Reviews weigh the channel as a whole.
- **Our approach:** every video has a distinct teaching idea, the instructor's own voice and opinions, original diagrams and real code. Templates give us consistency, but no two videos share the same script shape. Disclose realistic synthetic media where YouTube requires it.

> **Design consequence**
>
> The pipeline *accelerates* a human instructor; it doesn't replace one. The instructor writes or approves every script, records or clones **their own** voice, and signs off each lesson. That's also what makes it good.

04 Product: what we sell and publish

Course family (Udemy)

#	Course	Scope	Length
1	**AI Engineering for Forward Deployed Engineers** LAUNCH FIRST	LLM fundamentals → prompt and context engineering → RAG → tools / MCP → agents → evals → shipping into a customer's stack, plus the FDE delivery craft. Capstone included.	8–10 h
2	The Forward Deployed Engineer Playbook	Discovery, scoping, stakeholders, UAT, last-mile integration, ROI and executive reporting. The "consultant half" of the job.	4–6 h
3	Integration and Data Engineering for FDEs	APIs, auth, legacy systems, systems of record, pipelines, warehouses	8–10 h
4	Cloud and DevOps for FDEs	Docker, Kubernetes, Terraform, a cloud platform, observability, day-2 ops	6–8 h
★	The Complete FDE Bootcamp (bundle)	Courses 1–4 plus the foundations fast-track and full capstone	30–40 h

Why start with Course 1: it has the highest demand, it's where our animated explanations add the most, and it's less crowded than generic Python, SQL or cloud courses.

YouTube formats

Shorts · 3–5/week

One concept in 45–60 s ("What is RAG?", "MCP in 60s"). Like our API Short. Top of funnel.

Explainers · 1/week

8–14 min, 16:9. A deeper concept or a mini build. Often a cut-down course lesson.

Case files · 2/month

"A day as an FDE": a fictional customer problem solved end to end. Shows the brand's personality.

Funnel

Shorts → Explainers → Free lead magnet (our own FDE skills checklist PDF) → Email list → Udemy coupon launch → Bundle / own platform later

05 Curriculum architecture (our own structure)

The roadmap's ~100 skill areas are used privately as a coverage checklist. The module grouping, order, depth and teaching approach below are ours.

Module	Skill areas covered	Course
M0 · The FDE role	What FDEs do, role tiers and responsibilities, transition paths from SWE / data / consulting	All (free preview)
M1 · Engineering fast-track	Python, JS/TS, SQL, Git/GitHub, Linux, DSA and system design essentials	Bundle
M2 · Integration engineering	API design, REST/GraphQL, authentication, API security, Postgres and NoSQL, legacy systems, system of record, enterprise workflows	3
M3 · Data for FDEs	Data modeling, pipelines, Airflow, Spark, warehouses, Delta, privacy and compliance	3
M4 · Cloud and DevOps	Containers/Docker, Kubernetes, Terraform, AWS/GCP/Azure, observability, DevSecOps, day-2 operations	4
M5 · AI engineering	LLM fundamentals, choosing model providers, prompt and context engineering, prompt management, RAG, vector DBs, tools/functions, MCP, agent architectures, multi-agent, memory and state, debugging agents, eval pipelines, inference optimization, managed ML services, MLOps, AI governance	**1**
M6 · AI-assisted delivery	Coding agents and AI IDEs (e.g. Claude Code, Codex, Cursor), responsible "vibe coding", reviewing AI output	1
M7 · The FDE craft	Discovery and scoping, requirements, stakeholder management, presenting options, scope/speed/quality trade-offs, time-to-value, UAT, last-mile integration, product feedback loop, technical writing, executive reporting, ROI, productized consulting	1 (core) · 2 (deep)
Capstone	A fictional enterprise client engagement: discovery call → scoped proposal → AI agent integrated with a legacy system of record → eval suite → deployment → exec readout	1 · Bundle

Course 1 outline (draft, ≈ 45 lessons)

1. **Orientation**: the FDE role, the capstone client brief, setup
2. **LLMs for builders**: tokens, context windows, cost and latency, picking a provider
3. **Prompt and context engineering**: structure, few-shot, context budgeting, prompt versioning
4. **RAG that survives production**: chunking, embeddings, vector DBs, hybrid search, citations
5. **Tools, function calling and MCP**: connecting models to customer systems safely
6. **Agents**: architectures, memory and state, multi-agent, debugging agent logic
7. **Evals**: building eval pipelines, regression gates, LLM-as-judge pitfalls
8. **Shipping it**: inference optimization, observability, governance, security
9. **Delivery craft**: scoping, UAT, time-to-value, executive reporting, ROI
10. **Capstone**: the full engagement, end to end

Every coding lesson ships with a tagged commit in the course code repo, tested in CI.

06 Brand

Positioning

> **One-liner (draft)**
>
> "Learn to ship AI where it actually matters: inside the customer's systems." Practical, engineering-first, a little witty, never hype.

Name: decide in week 1

Requirements: short, ownable, not implying any affiliation with roadmap.sh or an employer, and the YouTube handle, .com/.dev domain and social handles are available. Avoid names close to existing players (e.g. "FDE Academy" is taken by fde.academy).

Candidate	Angle	Notes
Last Mile Engineer	The FDE's job is the last mile between AI and the customer	Memorable metaphor; check handle and domain
Shipside	Engineers who sit beside the customer and ship	Short, brandable; check conflicts
Deployed	Direct, role-adjacent	Generic, so the domain may be hard

Visual identity (extends the API Short style)

System

- Dark navy base (#0B0F1E) with neon accents: cyan = client/app, pink = server/system, yellow = API/AI layer, lime = success. **Colors carry meaning across every video.**
- Type: Inter (Black for titles, SemiBold for UI), JetBrains Mono for code.
- Motion language: snappy ease-out-back pops, 0.2 s white flash on cuts, punch-in zoom, packets moving along wires.

Brand kit deliverables

- Logo mark and wordmark, channel banner, avatar
- 2-second sonic and visual intro sting, outro end screen
- Lower thirds, chapter cards, code-window and terminal frames
- Thumbnail template set (3 layouts), caption style
- Voice and tone guide (do/don't examples)

The brand kit lives in the repo as code (Remotion components plus design tokens), so every video uses the same source of truth.

07 The production pipeline

One source of truth per lesson: a structured **lesson spec**. Everything else is generated from it, then checked by machines and by a human.

1 Research → 2 Lesson spec → 3 Code + tests → 4 Voice → 5 Scenes → 6 Audio mix → 7 Render variants → 8 QA gates → 9 Publish → 10 Learn

Stage	What happens	Tools
1 Research	Outline from the coverage checklist; collect primary sources; list the claims that need fact-checking	Claude, official docs, papers
2 Lesson spec	YAML/Markdown file: learning goal, beats, narration per beat, scene type per beat, code refs, on-screen text, quiz. The instructor approves it.	Git, Claude for drafts
3 Code + tests	Every snippet shown on screen comes from the course code repo and is run in CI. Lessons reference tagged commits.	GitHub Actions, pytest / vitest
4 Voice	The instructor records (preferred for intros and opinions) *or* TTS with a verified clone of **their own** voice. TTS returns **word timestamps** for exact captions (our Short had to estimate them).	ElevenLabs (with-timestamps API); mic + Audacity / Descript for recording
5 Scenes	A library of ~12 parameterized scene templates (below) driven by the spec	Remotion (React), Shiki for code highlighting
6 Audio	Music bed, SFX, ducking under voice, loudness to −14 LUFS	ffmpeg; own synthesized music or a licensed library
7 Render	16:9 lesson (1080p), 9:16 Shorts cut-downs, SRT captions, chapters, thumbnail, description and tags	Remotion CLI / Lambda, ffmpeg
8 QA gates	Automated checks plus a human review checklist (§8)	Scripts in CI; reviewer sign-off
9 Publish	Scheduled YouTube upload; Udemy uploaded manually (no public course-authoring API)	YouTube Data API, vidIQ for packaging
10 Learn	Retention curves, CTR, comments, Udemy reviews and Q&A feed the backlog	YouTube Studio, vidIQ, Udemy dashboard

Scene template library (v1 = first 5 in bold)

Title / hook card
Concept diagram (boxes, wires, packets)
Code walkthrough (typed reveal, line highlights)
Diff view

Terminal run (scripted with VHS)
Talking head + overlay
Comparison / table
Timeline / sequence

Quiz / pause-and-think
Architecture zoom
Metric / chart
Recap and next-up card

Example lesson spec

```yaml
# lessons/c1-m4-rag/04-hybrid-search.yaml
id: c1-m4-04
title: "Hybrid search: why vectors alone fail"
goal: "Explain and implement BM25 + vector fusion"
voice: instructor          # or: clone
beats:
  - scene: hook
    say: "Your RAG bot can't find invoice INV-2291. Here's why."
  - scene: diagram
    preset: query-to-two-indexes
    say: "Keyword search nails exact IDs; vectors nail meaning..."
  - scene: code
    ref: course-code@c1-m4-04:rag/hybrid.py#L12-L40
    highlight: [18, 27]
    say: "Reciprocal rank fusion merges both lists..."
  - scene: terminal
    tape: tapes/hybrid-demo.tape
quiz: [{q: "Why does BM25 find INV-2291?", a: "Exact token match"}]
shorts: [{from: hook, to: diagram, title: "Why your RAG can't find IDs"}]
```

Tooling choice: rendering engine

Option	Pros	Cons	Verdict
Remotion	React + CSS, huge ecosystem, captions package, live preview Player, cloud rendering	Paid company license once the company has more than 3 people	**Recommended**
Motion Canvas	MIT, great code animations, TypeScript	Smaller ecosystem, weaker for long-form layouts	Fallback
Manim	Beautiful math / diagram animation	Slow iteration, not built for UI or code	Special scenes only
Pillow + ffmpeg (our Short)	Zero dependencies, fast	Everything hand-drawn; doesn't scale to 45 lessons	Retire after MVP

08 Quality bar: definition of done

Automated gates (CI blocks the render)

- All lesson code passes tests; snippets match the tagged commit
- Resolution ≥ 1080p, 30 fps, 16:9 (Shorts 9:16)
- Loudness −14 LUFS ±1, true peak ≤ −1 dBTP; no clipping
- Captions present, word timing drift < 100 ms
- Spelling and terminology lint (glossary-enforced)
- No text outside safe areas; contrast check on captions
- Links in descriptions resolve

Human review (instructor signs off)

- Technically correct; claims sourced; nothing outdated
- One clear learning goal, met by the end
- Pacing: something changes on screen at least every 5–8 s in explainers
- The voice sounds natural; names pronounced correctly
- Original: no copied text, diagrams or roadmap reproduction
- Accessible: readable at phone size, colour-blind-safe meaning
- Would I pay for this? If not, it doesn't ship.

Lesson production targets (estimates to validate in the pilot)

Step	Human time per 10-min lesson	Notes
Spec review and edits	45–60 min	Claude drafts; the instructor sharpens the teaching
Code and tests	30–90 min	Varies with lesson type
Voice (record or approve TTS)	20–40 min	Clone + pickups is faster
Review render + fixes	30–45 min	Two passes
Total	**≈ 2–4 h**	vs. a typical 8–15 h for hand-edited animated lessons

09 Cost model

Prices are from vendor pages and third-party summaries as of Oct–Nov 2026. Verify before buying.

Item	Option	Cost	When
Voice (TTS + own voice clone)	ElevenLabs Creator: ~121 min/month, 1 professional voice clone (own voice, verified), commercial use	$22/mo	Pilot, YouTube cadence
	ElevenLabs Pro: ~600 min/month	$99/mo	Course production months (Course 1 ≈ 500–600 min incl. retakes)
	Cheaper alternatives: OpenAI TTS (≈ $0.015/min, some models scheduled for deprecation), Cartesia (≈ $0.035/min)	Pay-per-use	Drafts / previews only
Rendering framework	Remotion: free for individuals and companies ≤ 3 people	$0	Always (re-check if the team grows)
Rendering compute	Local machine or cloud session; Remotion Lambda optional	$0–10/mo	—
Music and SFX	Own synthesized (as in the Short) or a licensed library subscription	$0–20/mo	From launch
Microphone (if recording)	Dynamic USB mic + basic room treatment	$100–200 one-off	Week 1
Domain + email list	Domain; email tool free tier	~$15/yr; $0	Week 1
Research / scripting / analytics	Claude, vidIQ (existing subscriptions)	existing	—
Typical monthly total		**≈ $25–50 (normal months) · ≈ $100–150 (heavy production months)**	

Revenue mechanics (illustrative, not a forecast)

- Udemy courses often sell at discounted prices of ~$10–20. Organic sales pay the instructor the blended marketplace rate (check the current %), while **your own coupon links pay ~97%**.
- That's why YouTube matters: every buyer we bring ourselves is worth roughly 2–3× an organic buyer.
- At ≈ $150/month peak cost, break-even is a few dozen sales a month. The upside comes from the bundle, Udemy Business (needs high polish) and, later, our own platform.

10 Execution plan

Phase	Deliverables	Exit criteria
0 · Setup Week 1	New repo (structure below) · brand name + handles + domain · YouTube channel · fix ElevenLabs key (currently holds a key ID, not the key) · voice decision: record vs. own-voice clone · record a 30-min clean voice sample if cloning	Channel live, repo created, voice works end to end
1 · Pipeline MVP Weeks 2–3	Lesson-spec schema · 5 scene templates in Remotion · TTS + word timestamps → captions · audio mix + loudness · automated QA script · 1 pilot lesson (16:9) + 3 Shorts · Udemy test video submitted	Pilot passes all gates; Udemy feedback OK; you'd pay for it
2 · Course 1 production Weeks 4–9	~45 lessons in batches of 5–8 · capstone code repo with CI · remaining 7 templates · YouTube: 3 Shorts + 1 explainer per week · email list + lead-magnet PDF	All lessons pass the definition of done
3 · Launch Week 10	Udemy listing (with AI-tool disclosure), promo video, coupon launch to the email list and YouTube, launch-week content	Course live; first reviews
4 · Expand Month 3+	Course 2 (Playbook) → 3 → 4 → bundle · monthly updates from Q&A · consider own platform	Repeatable 1 course / 6–8 weeks

Proposed repo structure

```
fde-course/
├── brand/            # tokens, logo, fonts, intro sting, thumbnail templates
├── curriculum/       # coverage checklist (private), course outlines
├── lessons/          # one YAML lesson spec per lesson
├── course-code/      # runnable lesson code + capstone, tested in CI
├── studio/           # Remotion project: scene templates, compositions
├── pipeline/         # voice, captions, audio mix, QA, render, publish scripts
├── tapes/            # VHS terminal recordings
├── assets/           # music, sfx, voice takes (large files via Git LFS)
└── docs/             # this blueprint, style guide, review checklist
```

11 Decisions needed and risks

Decisions for you (in priority order)

#	Decision	My recommendation
1	Your voice: record, clone or both?	**Both:** record intros and opinions; use a verified clone of your own voice for long technical narration. Best for authenticity and speed.
2	On camera?	Face in the course intro, the promo and the "case files" videos is enough; animated lessons need no camera.
3	Brand name	Pick from §6 after availability checks; decide in week 1.
4	First course scope	Course 1, "AI Engineering for FDEs", 8–10 h.
5	Monthly budget ceiling	$150/month during production.
6	Contact roadmap.sh's author?	Yes, optional upside: a friendly note about the course, no dependency.

Risks and mitigations

Risk	Mitigation
Content seen as "AI slop" (Udemy quality flags, YouTube demonetization)	Human-led scripts, instructor voice, original diagrams, runnable code, varied formats, AI disclosure
Copyright: roadmap or third-party material	Follow skill areas only; originality check in human review; licensed or own music
Fast-moving AI field makes lessons stale	Specs make re-renders cheap; quarterly refresh; pin tool versions in course code
Crowded "FDE" keyword on Udemy	Differentiate on depth, projects and quality; build our own audience on YouTube first
Vendor price or policy changes (TTS, Remotion, Udemy)	Abstract the voice provider behind one interface; keep Motion Canvas as fallback; email list as an owned channel
Scope creep (trying to cover all 100 topics at once)	One course at a time; Course 1 first; strict definition of done
Burnout from cadence	Batch production; Shorts cut automatically from lessons

12 Sources

Accessed 13 Nov 2026. Third-party summaries are marked; verify prices and policies on official pages before relying on them.

- roadmap.sh repository license (personal use only; publishing requires consent): github.com/kamranahmedse/developer-roadmap/blob/master/license (now nilbuild/developer-roadmap)
- Udemy, Course Quality Checklist and Video Standards: support.udemy.com/hc/articles/229232767-Video-Standards; support.udemy.com/hc/en-us/articles/229604988
- Udemy, AI use in course content: support.udemy.com/hc/en-us/articles/30999984483607
- Udemy, subscription revenue-share update: teach.udemy.com/enabling-investment-subscription-terms-update/
- Udemy community, updated business terms (blended rate): community.udemy.com/en/discussion/79320
- S&P Global, Coursera–Udemy deal (Sep 2026): spglobal.com/market-intelligence/en/news-insights/research/2026/09/
- YouTube inauthentic-content policy coverage: techrepublic.com/article/news-youtube-ai-video-monetization-rules; metricool.com/youtube-ai-content-rules-monetization/; thedesk.net/2026/07/youtube-clamps-down-low-quality-ai-slop/ (third-party)
- FDE demand and pay: paraform.com/insights/forward-deployed-engineer-demand-quadrupled; leaddev.com/hiring/the-rise-of-the-forward-deployed-engineer; analyticsindiamag.com (India 19×); bettsrecruiting.com/blog/top-compensation-trends-for-forward-deployment-engineers-in-2026/ (third-party)
- Competition: udemy.com/course/forward-deployed-engineer-mastery/; udemy.com/course/forward-deployed-engineer-fde/; udemy.com/course/forward-deployed-engineer-the-complete-crash-course/; maven.com/rajat-dandekar/ai-fde-cohort; educative.io/courses/forward-deployed-engineer; fde.academy
- ElevenLabs pricing summaries: bigvu.tv/blog/elevenlabs-pricing-2026-plans-credits-commercial-rights-api-costs/; magichour.ai/blog/elevenlabs-pricing (third-party; official: elevenlabs.io/pricing)
- TTS price comparisons: texttolab.com/blog/openai-tts-pricing; gradually.ai (gpt-4o-mini-tts vs Cartesia) (third-party)
- Remotion license: remotion.dev/docs/license; remotion.dev/blog/company-licenses

> **Next step**
>
> Answer the 6 decisions in §11 (even "go with your recommendations" works). Once the new repo exists, Phase 0 starts and I'll turn §7 into a step-by-step implementation plan.